What is Apache Airflow?

Apache Airflow is an open-source tool to programmatically author, schedule, and monitor workflows. It is used by Data Engineers for orchestrating workflows or pipelines. One can easily visualize your data pipelines' dependencies, progress, logs, code, trigger tasks, and success status. Complex data pipelines are managed using it.

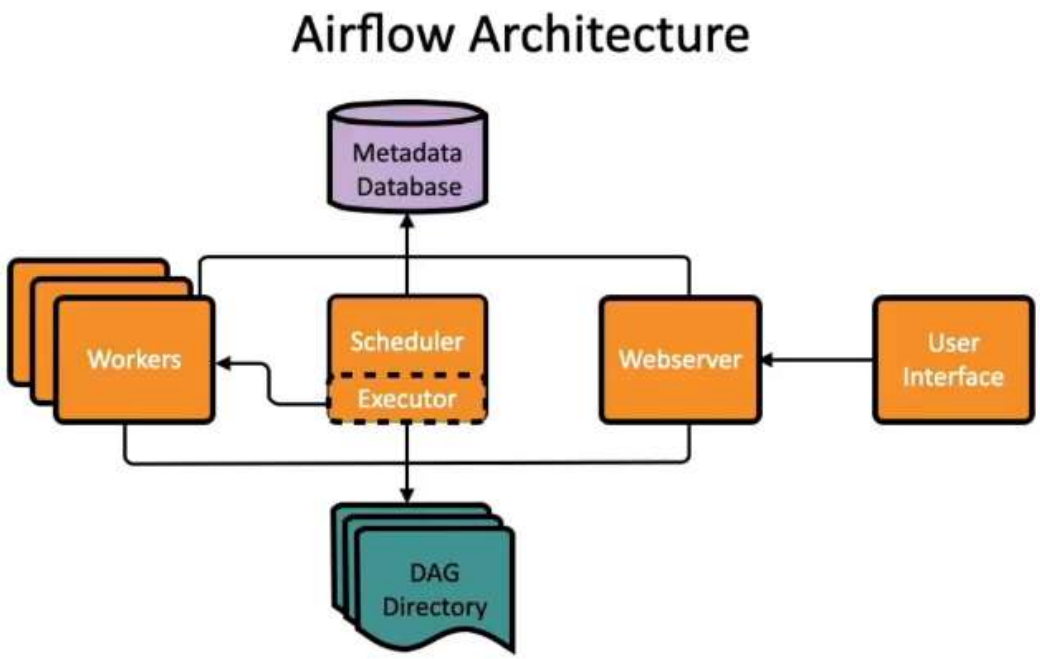
- These data pipelines are used to deliver datasets that are easily used by business intelligence applications or machine learning models where a huge amount of data is required.
- It is one of the most robust platforms for data engineers.
- Batch-oriented workflows are developed, scheduled, and monitored efficiently. Apache Airflow is a workflow engine that easily schedules and runs complex data pipelines.

Advantages of Apache Airflow:

- The Airflow community is very large and is still growing. It was started back in 2015 by Airbnb. So, there's a lot of support available.
- Apache Airflow is highly extensible which allows it to suit any environment. Custom cases are implemented very easily.
- The pipelines are generated dynamically and are configured as code using Python programming language.
- Rich scheduling and execution semantics are used to easily define complex pipelines and keep them running at regular intervals.
- With a little bit of Python knowledge, one can go about deploying on Airflow.

- It is free and open-source and has a lot of active users.
- Users can monitor and manage their workflows.

Airflow architecture:



1. DAG (Directed Acyclic Graph)

A DAG is the workflow definition in Apache Airflow written using Python code. It defines tasks, their execution order, dependencies, schedules, and retry rules. DAGs act as the blueprint for how a workflow should run from start to finish.

2. Scheduler

The Scheduler is the core component responsible for monitoring DAGs and deciding when tasks should execute. It continuously checks schedules, task

dependencies, and execution conditions before triggering jobs. The scheduler ensures workflows run automatically and in the correct sequence.

3. Metadata Database

The Metadata Database stores all operational information related to Airflow workflows and task executions. It maintains DAG states, task statuses, logs references, user information, and execution history. Airflow relies heavily on this database for coordination between components.

4. Executor

The Executor determines how and where tasks are executed after being scheduled. It receives tasks from the scheduler and distributes them to worker processes or machines. Different executors such as LocalExecutor, CeleryExecutor and KubernetesExecutor support different scalability requirements.

5. Workers

Workers are the systems or processes that actually perform the task execution. They run scripts, SQL queries, API calls, Spark jobs, or other operations defined in DAG tasks. Multiple workers can run tasks in parallel to improve performance and scalability.

6. Web Server (UI)

The Web Server provides the graphical user interface for managing and monitoring Airflow workflows. It allows users to visualize DAGs, track task execution, view logs, and retry failed tasks. The UI improves operational visibility and simplifies workflow management.

What is DAG?

DAG (Directed Acyclic Graph)

A DAG in Apache Airflow is a workflow definition that organizes tasks and their execution order. It represents a sequence of tasks where each task depends on another task, and the workflow moves in one direction without looping back.

The term:

- **Directed** → tasks follow a specific direction/order
- **Acyclic** → no circular loops allowed
- **Graph** → tasks are connected like a flowchart

What are Executors?

An Executor in Apache Airflow is the component responsible for deciding how and where tasks run after the scheduler triggers them. It acts as the bridge between the scheduler and the workers executing the tasks.

The executor manages:

- Task distribution
- Parallel execution
- Resource utilization
- Scalability

Different executors are used based on system size and workload.